CAS (Heimdall) – design basis for control

Parameters, dynamic model and linear model

Design basis version 2.3.11

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1 Purpose

This document sets out the parameters and the dynamic model used to design the level and oxygen control for the ScaleAQ closed aquaculture system (CAS), PLC project Heimdall. Every parameter has a value, a unit, a source and a reason for the choice. It replaces the parameter set in *CAS model.tex*; Section 10 lists the changes.

One source of truth. The parameters live in `design_basis.toml` [TOML] and the equations in `casdesign/design.py` [MODEL], both in `workfolder-python/CasDesignBasis`. Two simulators run them (Section 8): CasRioSim, the remote I/O over Modbus TCP and the pen physics, and CasFramoSim, the Framo inlet pump controllers over OPC UA. Their version is the design basis version (2.3.11). The tables, matrices and difference equations in this document are generated from the same files by `tools/design_report.py` [PY] into `generated/`. To change the design:

1. change `design_basis.toml` (or `design.py`) and bump `meta.version`;
2. run `python tools/selftest_plant.py`, which checks that the model linearizes to exactly the matrices in Section 6;
3. run `python tools/design_report.py` and rebuild this document;
4. push to `main`. CI publishes both simulator images with the new version.

Values marked [A] are assumptions that are not yet supported by a document. They are listed in Section 11 and must be confirmed.

Sources

ID	Document
[DB]	10925-CAS-001-02 Design basis, rev. 2, Automation (working document)
[FD]	P-15169-CAS-FD-01 Funksjonsbeskrivelse CAS, rev. B incl. tracked changes up to 28.09.26
[SPEC]	Framo 0000_CAS_SPEC_CTRLSYS, Kontrollsystem (prototype), rev. B, 08.10.2025
[SR2000]	Framo SR2000 pump specification (datasheet)
[PF]	pumpeflow_Sr2000_CAS.xlsx (pump power and system head reference)
[O2S]	Beregningsark dimensjonering ved bortfall av pumper eller strøm, 17.03.26
[CT]	Oppsett innløsere CT.xlsx (Chucaotech oxygenator sizing)
[CTE]	CT002-CS-022 v0-2, oxygen transfer efficiency test, 23.07.2024
[B2865]	Bürkert 2865 proportional valve, datasheet
[S418]	SUTO S418 flow meter, datasheet
[RDO]	RDO Core (Orbit 864) dissolved oxygen sensor, spec sheet
[VEGA]	VEGAPULS C 21 radar level sensor, specification sheet
[LTU]	LTU801 pressure level transmitter, IOM
[SO]	Sensor Overview CAS V1.xlsx
[B8605]	Bürkert 8605 PWM driver, Linde default setup, 22.02.22
[W455]	WAGO 750-455 4AI 4–20 mA product manual, 2453453963, valid from HW 07 / FW 04
[RDOM]	In-Situ RDO Core Digital Interface Specification (Modbus register map)
[W652]	WAGO 750-652 serial interface RS-232/RS-485 manual, version 2.0.0
[HEIM]	Heimdall CODESYS project (scaleaq/heimdall-plc): ModbusRTUData, GVL_CONSTANT
[OGA]	K. Ogata, *Modern Control Engineering*, 5th ed., Prentice Hall, Upper Saddle River, NJ, 2010, ISBN 978-0-13-615673-4
[PRO]	J. G. Proakis and D. G. Manolakis, *Digital Signal Processing*, 4th ed., Pearson Prentice Hall, Upper Saddle River, NJ, 2007
[UPID]	CODESYS Util library, function block `PID` (library documentation)
[TOML]	CasDesignBasis/design_basis.toml, version 2.3.11 (all parameters)
[MODEL]	CasDesignBasis/casdesign/design.py (model equations, used by the simulators)
[PY]	CasDesignBasis/tools/design_report.py (generates this document's tables)

2 System and control structure

Water is pumped into the pen by three Framo SR2000 inlet pumps through flexible tunnels taken in at 10–30 m depth. It leaves through a bottom centre outlet that has four hydraulic dampers. The pen level is held a few centimetres above the outside sea level (the *over-height* Δh), and that over-height drives the outflow. Oxygen is dissolved in a side stream on each inlet (oxy-pump and

Chucaotech oxygenator), with one oxygen gas flow loop per line.

The control structure follows [FD] §3.2.2–3.2.4 (Figure 1):

- **LIC01**, level: PI on Δh_{avg}. It sends one common speed setpoint to all running inlet pumps. The dampers are held at a fixed opening per operating mode.
- **OIC01**, oxygen (master): PI on the pen O_2 concentration in mg/L. [FD] names OT03 at the outlet, with fallback to the lowest remaining sensor; Heimdall uses the mean of the inside sensors (open item). Its output is the total O_2 flow to the pen, shared between the lines in auto (Section 7.2); each share is the setpoint of that line's flow loop.
- **FIC41–43**, O_2 flow (slave): PI on the gas flow (FT41–43). It drives the PWM proportional valve OV41–43.
- **PIC41–43**, oxy pump: PI on the pressure after the oxy pump. It sets the pump's VSD (ATV320) speed to hold 4.0 bar absolute into the oxygenator.

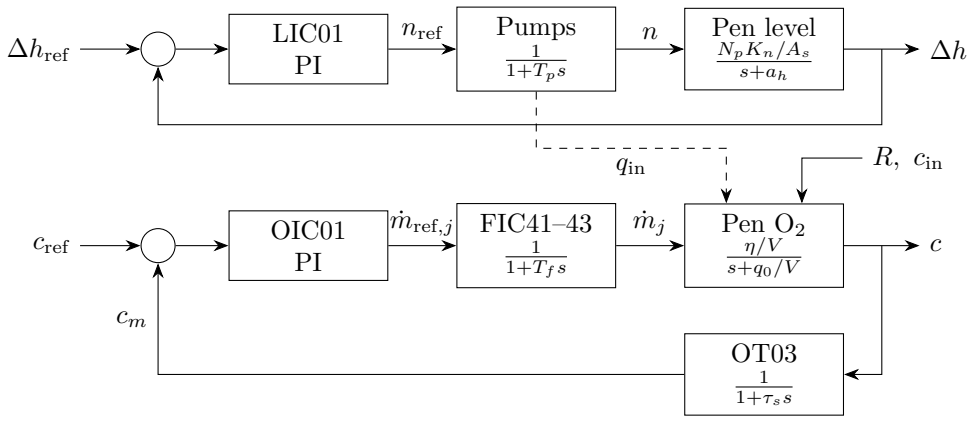


Figure 1: Control structure. The dashed line is the coupling from pump flow to oxygen.

3 Parameters

All tables in this section are generated from [TOML]; each row carries its own source and reason there.

3.1 Pen and outlet

Symbol	Value	Unit	Source	Reason
V	20 000	m^3	[DB] 1.2	Design volume.
D	50.5	m	[DB] tab. 7	Tarpaulin diameter. The free surface $A_s = \pi D^2/4$ sets the level capacitance; the surface is treated as rigid.
D_o	17.3	m	[DB] tab. 7	Tarpaulin depth, taken as the outlet depth.
ρ	1025	kg/m^3	[DB] tab. 21	Sea water density.
$\Delta\rho$	2	kg/m^3	[DB] 4.4.1	Density difference [DB] uses for the inner water height. Inside water from 10–30 m is assumed denser, giving the extra outlet head $h_\rho = D_o\Delta\rho/\rho$.[A]
full-open mode	O-45	–	[FD] tab. 10, [DB] tab. 16	Outlet conductance at full damper opening, taken from O-45. O-35 and O-45 give the same conductance (30.8 vs. 30.9), consistent with a fully open outlet. Conductance is assumed linear in the opening.[A]
A_s	2003	m^2	$\pi D^2/4$	Free surface area.
h_ρ	3.38	cm	$D_o\Delta\rho/\rho$	Extra driving head at the outlet.
$k_{d,\text{full}}$	30.94	$m^{2.5}/s$	$Q_0/\sqrt{\Delta h_0 + h_\rho}$	Outlet conductance at full opening (O-45).

3.2 Inlet pumps

Symbol	Value	Unit	Source	Reason
Q_{rated}	20 000	m^3/h	[SR2000]	Rated flow at rated speed.
$n_{\min}$	40	rpm	[SR2000], [FD] tab. 8	Minimum speed, also the LIC01 output range.
$n_{\max}$	130	rpm	[SR2000], [FD] tab. 8	Maximum speed. $K_n = Q_{\text{rated}}/n_{\max}$ by the affinity law $Q \propto n$, which holds because the system head is almost purely $\propto Q^2$ [PF] and the static over-height is below 10 cm.
N	3	–	[DB] 3.4	Three inlets, each with one SR2000 and one oxygenator.
T_p	10	s	-	Time constant from speed setpoint to actual speed (VSD ramp and water column). To be confirmed by Framo.[A]
$P_{\max}$	60	kW	[SR2000]	Motor power at $n_{\max}$. Power follows the affinity law $P = P_{\max}(n/n_{\max})^3$; at 130 rpm this gives the datasheet torque of 4407 Nm.
z_p	72	–	[SR2000]	Motor poles. Electrical frequency $f = nz_p/120$, 78 Hz at 130 rpm.
I_{rated}	110	A	[SR2000]	Rated current at 400 V. Current is scaled from power and voltage.
U_{rated}	400	V	[SR2000]	Rated voltage. Motor voltage is taken proportional to frequency (V/f).
K_n	0.042735	$(\text{m}^3/\text{s})/\text{rpm}$	$Q_{\text{rated}}/n_{\max}$	Flow per rpm.

3.3 Operating modes

Flow, level setpoint and number of pumps per mode come from [FD] tab. 10 and [DB] tab. 16. The other columns follow from the model: $n_0 = Q_0/(N_p K_n)$, $k_d = Q_0/\sqrt{\Delta h_0 + h_\rho}$, the damper opening $k_d/k_{d,\text{full}}$, and the level time constant $1/a_h$ (Section 6).

Mode	Q_0 [m^3/h]	Δh_0 [cm]	N_p	n_0 [rpm]	k_d [$\text{m}^{2.5}/\text{s}$]	Opening [%]	$1/a_h$ [s]
O-35	34 000	6.0	3	73.7	30.84	100	39.8
O-45	27 000	2.5	2	87.8	30.94	100	31.4
O-60	20 000	0.5	2	65.0	28.22	91	27.9
S-75	16 000	0.5	2	52.0	22.58	73	34.9

O-45 is the nominal design point. It is the [DB] dimensioning case (case 3: 1 000 t, 27 000 m^3/h,

45 min), and the mode that must be held with one pump out. The linear model and the discrete equations are given for O-45 with 2 pumps.

3.4 Water and oxygen

Symbol	Value	Unit	Source	Reason
T_w	10	°C	[DB] tab. 21	Design temperature.
S_w	34	ppt	[O2S]	Salinity.
$c_{\text{in}}/c_{\text{sat}}$	0.8	–	[O2S]	Intake water at 80 % O_2 saturation; saturation by the Weiss formula of [O2S].
$c_{\text{ref}}/c_{\text{sat}}$	0.9	–	[O2S]	Outlet design criterion, 90 % saturation. [FD] has no O_2 setpoint yet.[A]

Symbol	Value	Unit	Source	Reason
$\dot{m}_{\max}$	104.2	kg/h	[DB] tab. 13, [CT]	Maximum O_2 gas flow per oxygenator (CT10, sea water, 15 % gas/liquid ratio). An oxygenator adds O_2 only while its own inlet pump runs.
η	0.8	–	[CTE]	Transfer efficiency; the test gave 0.81 initial efficiency, rounded down.[A]
T_f	2	s	design choice	Closed-loop time constant of FIC41-43, used in the control design model.
τ_v	0.22	s	[B2865], [S418]	Open-loop lag of valve and flow meter, used in the simulator: valve < 25 ms, flow meter $t_{90} = 0.5$ s ($\tau_v = t_{90}/\ln 10$).
u_{cut}	2	%	[B8605]	Driver cut-off; below this the valve is closed (Linde default). Above it the gas flow is linear in the command up to $\dot{m}_{\max}$.
$u_{\%}$	4-20 mA	–	user, [B8605]	4 mA (0 %) = fully closed, 20 mA (100 %) = fully open. The PLC writes 0–65535 for 0–100 % to the 750-563 output (factory setting: 4–20 mA over the full 16-bit word); the Bürkert 8605 driver takes the 4–20 mA (Linde default) and sets the coil current. Below the cut-off the valve stays closed, so a lost signal is expected to close the valve (no O_2 on that line).
τ_s	19.54	s	[RDO]	O_2 sensor as first order fitted to $T_{90} < 45$ s ($\tau_s = T_{90}/\ln 10$). [RDO] gives $T_{63} < 5$ s, so this is conservative.
c_{sat}	9.09	g/m^3	[O2S]	Saturation at T_w, S_w (Weiss).
c_{in}	7.27	g/m^3		O_2 in the intake water.
c_{ref}	8.18	g/m^3		O_2 setpoint.
$\dot{m}_{\max}$	28.94	g/s		Maximum O_2 gas flow per line.

T_f is used in the control design model, where FIC41–43 is closed. τ_v and the cut-off are used in the simulator, where the PLC's FIC drives the valve itself (Section 8).

3.5 Fish

Symbol	Value	Unit	Source	Reason
ρ_f	50	kg/m^3	[DB] 1.2	Maximum density; biomass $B = \rho_f V$.
W	1000	g	[DB] tab. 18	Average weight, [DB] case 3.
U	41	cm/s	[O2S]	Recommended swimming speed for 1 kg at 10 °C.
B	1 000 000	kg	$\rho_f V$	Maximum biomass.
r	3.26	mg/(kg min)	[O2S]	Average post-smolt salmon, $r = 1.92\, W^{-0.27} T^{0.63} 10^{0.01U}$ ($= 196$ mg/(kg h)).
R_0	54.3	g/s	rB	Nominal total O_2 consumption of the fish.

3.6 Level measurement

Symbol	Value	Unit	Source	Reason
LT01–03	2	mm	[VEGA], [SO]	Radar, deviation ≤ 2 mm.
LT21–23	4	mm	[LTU], [SO]	Pressure transmitter, 0.2 % of an assumed 2 m full scale.[A]
σ_{LT0x}	1	mm	[VEGA], [PRO] 1.4.4	Standard deviation of the inside level reading, taken as half the ±2 mm deviation. The simulator adds it as white noise. It also dithers the 0.98 mm quantization, which the uniform quantization-noise model ($\Delta^2/12$) assumes; without it the PI hunts across one step.[A]
σ_{LT2x}	2	mm	[LTU]	Standard deviation of the outside level reading, half its ±4 mm deviation. Waves are not included.[A]
h_{span}	2	m	[VEGA], [W455], user	Span of the 4–20 mA level signal. The 750-455 puts the value in bits 14–4 of the word, 0–2047 (20 mA = 0x7FF0; bit 15 is the sign, bits 3–2 reserved, 1–0 diagnostics) [W455], confirmed by WAGO 2026-10-07. One step is 2000/2047 = 0.98 mm, inside the ±2 mm sensor deviation; the over-height, a difference of two readings, moves in steps of about 1 mm.
raw 0	low	–	[VEGA]	Which end of the span reads 0 (4 mA). "low": 0 = lowest water level in the span (VEGA level-proportional output, 4 mA = min. adjustment). "high": 0 = highest level, as when the output follows the measured distance from the antenna. Not stated in the datasheets; read the min./max. adjustment and current output mode in the sensor.[A]
$h_{\mathrm{out},0}$	1	m	-	Outside sea level within the span, mid-range. Set by the sensor mounting height.[A]
OT01–03, OT21–22	5	–	[RDOM], [W652], [HEIM]	In-Situ RDO Core (Orbit 864), Modbus RTU on the RS-485 line of the 750-652, the first module on CommonRIO (Modbus words 0–11). Three inside the pen

The PV range of LIC01 is 0–20 cm [FD] tab. 8. The PV is formed per inlet pair, as [FD] §3.2.2 asks: $\Delta h_i = h_{\mathrm{LT0}i} - h_{\mathrm{LT2}i}$ from the scaled readings, filtered per pair (Section 7.3), then $\Delta h = \frac{1}{|V|}\sum_{i\in V}\Delta h_i$ over the set V of valid pairs. A pair is valid while both its transmitters are healthy (750-455 status bits 1–0 = 00) and enabled and the pair is enabled; if either transmitter is lost, the whole pair drops out, so a local offset of the remaining transmitter cannot leak into the PV. With no valid pair, LIC01 holds the pumps at a configured safe speed (Section 7).

4 Dynamic model

4.1 States, inputs and disturbances

$x_1 = \Delta h$ [m], over-height
$x_2 = n$ [rpm], actual pump speed
$x_3 = c$ [g/m^3], O_2 concentration in pen
$x_4 = \dot{m}_1$ [g/s], O_2 gas flow line 1
$x_5 = \dot{m}_2$ [g/s], line 2
$x_6 = \dot{m}_3$ [g/s], line 3
$x_7 = c_m$ [g/m^3], measured O_2 (OT03)

$u_1 = n_{\mathrm{ref}}$ [rpm], common pump speed setpoint
$u_2 = \dot{m}_{\mathrm{ref},1}$ [g/s], O_2 flow setpoint line 1
$u_3 = \dot{m}_{\mathrm{ref},2}$ [g/s], line 2
$u_4 = \dot{m}_{\mathrm{ref},3}$ [g/s], line 3
$d_1 = R$ [g/s], O_2 consumption of fish
$d_2 = c_{\mathrm{in}}$ [g/m^3], O_2 in intake water

The O_2 inputs are setpoints to the flow loops, not valve positions, because [FD] rev. B puts FIC41–43 in cascade under OIC01. To go from the [FD] valve scaling (0–100 %) to flow, use $\dot{m}_{\mathrm{ref}} = \dot{m}_{\max} \cdot u_{\%}/100$.

4.2 Water balance

The inflow is the total pump flow. The outflow goes through the damper opening, driven by the over-height plus the density head:

$$q_{\mathrm{in}} = N_p K_n \, n \tag{1}$$

$$q_{\mathrm{out}} = k_d \sqrt{\Delta h + h_\rho} \tag{2}$$

$$A_s \, \dot{x}_1 = N_p K_n \, x_2 - k_d \sqrt{x_1 + h_\rho} \tag{3}$$

$$T_p \, \dot{x}_2 = -x_2 + u_1 \tag{4}$$

Here k_d is the outlet conductance at the fixed damper opening of the mode. It is set so that the mode's flow Q_0 gives the mode's over-height Δh_0: $k_d = Q_0/\sqrt{\Delta h_0 + h_\rho}$.

The outlet follows the orifice law ($q \propto \sqrt{h}$), not a linear resistance. That halves the slope compared with a linear outlet, and makes the level time constant twice as long. This is the turbulent liquid-level system of [OGA] §4-2 ($Q = K\sqrt{H}$, eq. 4-1): its linearized resistance $R_t = 2\bar{H}/\bar{Q}$ is the slope used for a_h in Section 6, and the pen area A_s is the capacitance.

4.3 Oxygen balance

The pen is treated as one fully mixed volume (CSTR). Since $\frac{d}{dt}(Vc) = q_{\text{in}}c_{\text{in}} - q_{\text{out}}c + \eta\sum \dot{m}_j - R$ and $\dot{V} = q_{\text{in}} - q_{\text{out}}$:

$$V\,\dot{x}_3 = N_p K_n\,x_2\,(d_2 - x_3) + \eta\,(\sigma_1 x_4 + \sigma_2 x_5 + \sigma_3 x_6) - d_1 \tag{5}$$

$$T_f\,\dot{x}_{3+j} = -x_{3+j} + u_{1+j}, \qquad j = 1,2,3 \tag{6}$$

$$\tau_s\,\dot{x}_7 = -x_7 + x_3 \tag{7}$$

The oxygenators are additive. The intake water brings its own oxygen through the term $q_{\text{in}}c_{\text{in}}$, and each oxygenator adds $\eta\dot{m}_j$ on top of that. The side stream through each oxygenator (288–486 m^3/h [DB] tab. 13) is only 2–4 % of a pump's flow and goes back into the same inlet, so it does not appear in the water balance.

$\sigma_j \in \{0,1\}$ says whether oxygen line j is in operation, and $N_o = \sum_j \sigma_j$ is the number of active lines. The oxygenator on a stopped inlet cannot add oxygen to the pen (confirmed; see also the interlock in [FD] tab. 2). Therefore $\sigma_j = 1$ only while inlet pump j runs, and $N_o = N_p$.

Equation (5) is exact for a fully mixed volume; no $\dot{V}$ term is lost. The water flow therefore acts on oxygen as $q_{\text{in}}(c_{\text{in}} - c)$. The intake water contains less oxygen than the setpoint ($c_{\text{in}} < c_{\text{ref}}$), so **more pump flow lowers the pen $\mathbf{O_2}$** while the fish consume oxygen and injection keeps c above c_{in}.

5 Operating points

At steady state $\Delta h = \Delta h_0$ and $c = c_{\text{ref}}$, and the N_o active lines share the load equally:

$$n_0 = \frac{Q_0}{N_p K_n}, \qquad \dot{m}_{j,0} = \frac{1}{N_o \eta}\Big(R_0 - Q_0\,(c_{\text{in}} - c_{\text{ref}})\Big) \tag{8}$$

Mode	V/Q_0 [min]	$N_p = N_o$	n_0 [rpm]	$\dot{m}_{j,0}$ [g/s]	Load of oxygenator [%]
O-35	35.3	3	73.7	26.22	90.6
O-45	44.4	3	58.5	25.48	88.0
O-45	44.4	2	87.8	38.22	**132.1**
O-60	60.0	3	43.3	24.75	85.5
O-60	60.0	2	65.0	37.12	**128.2**
S-75	75.0	2	52.0	36.49	**126.1**

The speeds in the table are inside 40–130 rpm. S-75 on three pumps would need 34.7 rpm, which is below the minimum speed, so S-75 can only run on two pumps. The oxygen requirement, not the water flow, decides how many pumps must run. At full biomass **three pumps must run in O-35, O-45 and O-60** (at reduced speed in O-45 and O-60). S-75 cannot hold c_{ref} at full biomass, and with two pumps $c = c_{\text{ref}}$ cannot be reached in any mode (Section 9). The linear model in the next section uses O-45 with two pumps, the redundancy case; around a 3-pump point only N_p and n_0 change.

6 Linear model

Linearize around the operating point (x_0, u_0, d_0) by a first-order Taylor expansion ([OGA] §2-7, eqs. 2-42–2-46) and write $\delta x = x - x_0$. The model holds for small deviations from the operating point; the simulators run the nonlinear equations. Define

$$a_h = \frac{\partial q_{\text{out}}}{\partial \Delta h}\frac{1}{A_s} = \frac{Q_0}{2A_s(\Delta h_0 + h_\rho)}, \qquad a_c = \frac{Q_0}{V}, \qquad b_c = \frac{N_p K_n (c_{\text{in}} - c_{\text{ref}})}{V} \tag{9}$$

Then

$$\delta\dot{\mathbf{x}} = \mathbf{A}\,\delta\mathbf{x} + \mathbf{B}\,\delta\mathbf{u} + \mathbf{E}\,\delta\mathbf{d}, \qquad \mathbf{y} = \mathbf{C}\,\delta\mathbf{x} \tag{10}$$

$$\mathbf{A} = \begin{bmatrix} -a_h & \frac{N_p K_n}{A_s} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{T_p} & 0 & 0 & 0 & 0 & 0 \\ 0 & b_c & -a_c & \frac{\eta\sigma_1}{V} & \frac{\eta\sigma_2}{V} & \frac{\eta\sigma_3}{V} & 0 \\ 0 & 0 & 0 & -\frac{1}{T_f} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{1}{T_f} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{1}{T_f} & 0 \\ 0 & 0 & \frac{1}{\tau_s} & 0 & 0 & 0 & -\frac{1}{\tau_s} \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ \frac{1}{T_p} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & \frac{\sigma_1}{T_f} & 0 & 0 \\ 0 & 0 & \frac{\sigma_2}{T_f} & 0 \\ 0 & 0 & 0 & \frac{\sigma_3}{T_f} \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{E} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ -\frac{1}{V} & a_c \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \tag{11}$$

The outputs are Δh_{avg}, the pump speed read back (MotorVelocity [SPEC]), OT03 and FT41–43:

$$\mathbf{C} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix} \tag{12}$$

Numerical values, O-45 with 2 pumps

Coefficient	Value	Unit
a_h	0.0319	1/s $\quad (1/a_h = 31.4\,\text{s})$
$N_p K_n / A_s$	4.2672×10^{-5}	$\text{m}/(\text{s}\,\text{rpm})$
$1/T_p$	0.1000	1/s
a_c	3.7500×10^{-4}	1/s $\quad (1/a_c = 44.4\,\text{min})$
b_c	-3.8835×10^{-6}	$\text{g}/(\text{m}^3\,\text{s}\,\text{rpm})$
η/V	4.0000×10^{-5}	$1/\text{m}^3 \quad$ (times σ_j)
$1/T_f$	0.5000	1/s
$1/\tau_s$	0.0512	1/s
$1/V$	5.0000×10^{-5}	$1/\text{m}^3$

6.1 Transfer functions

The matrices are triangular, so the transfer functions are products of first-order terms ([OGA] §5-2):

$$\frac{\Delta h}{n_{\text{ref}}}(s) = \frac{N_p K_n / A_s}{(s + a_h)(1 + T_p s)} \qquad K = 1.339 \text{ mm/rpm} \tag{13}$$

$$\frac{c_m}{\dot{m}_{\text{ref},j}}(s) = \frac{\eta / V}{(s + a_c)(1 + T_f s)(1 + \tau_s s)} \qquad K = \frac{\eta}{Q_0} = 0.1067 \ \tfrac{\text{g/m}^3}{\text{g/s}} \tag{14}$$

$$\frac{c_m}{R}(s) = \frac{-1/V}{(s + a_c)(1 + \tau_s s)} \qquad K = -\frac{1}{Q_0} = -0.1333 \ \tfrac{\text{g/m}^3}{\text{g/s}} \tag{15}$$

$$\frac{c_m}{n_{\text{ref}}}(s) = \frac{b_c}{(s + a_c)(1 + T_p s)(1 + \tau_s s)} \qquad K = \frac{N_p K_n (c_{\text{in}} - c_{\text{ref}})}{Q_0} = -0.0104 \ \tfrac{\text{g/m}^3}{\text{rpm}} \tag{16}$$

$$\frac{c_m}{c_{\text{in}}}(s) = \frac{a_c}{(s + a_c)(1 + \tau_s s)} \qquad K = 1 \tag{17}$$

The steady-state gains are for O-45 and are computed by [PY].

6.2 Discretization

Each first-order term $\frac{1}{1+\tau s}$ discretizes exactly with zero-order hold (ZOH) to $\frac{1-p}{z-p}$, where $p = e^{-T_s/\tau}$. With $T_s = 1\,\text{s}$ at O-45:

Term	τ [s]	p
Level, $1/a_h$	31.4	0.968638
Pump speed, T_p	10.0	0.904837
Pen O_2, $1/a_c$	2666.7	0.999625
O_2 flow loop, T_f	2.0	0.606531
O_2 sensor, τ_s	19.5	0.950110

All the equations below are for O-45 with 2 pumps running and $T_s = 1\,\text{s}$. They are in deviation variables (the δ is dropped), and the inputs are held constant over each sample.

State update

The solution of $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$ is $\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}(0) + \int_0^t e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}(\tau)\,d\tau$ ([OGA] §9-4, eq. 9-42). With the input held constant over a sample (zero-order hold) this gives the exact ZOH of the full state-space model, $\mathbf{x}_k = \mathbf{A}_d\mathbf{x}_{k-1} + \mathbf{B}_d\mathbf{u}_{k-1} + \mathbf{E}_d\mathbf{d}_{k-1}$, with $\mathbf{A}_d = e^{\mathbf{A}T_s}$ and $\mathbf{B}_d = \int_0^{T_s} e^{\mathbf{A}\lambda}\,d\lambda\,\mathbf{B}$ (likewise

$\mathbf{E}_d$). Row by row:

$$
\begin{aligned}
\Delta h_k &= 0.968638\,\Delta h_{k-1} + 3.995\,688\times10^{-5}\,n_{k-1} + 2.042\,285\times10^{-6}\,n_{\mathrm{ref},k-1}\\
n_k &= 0.904837\,n_{k-1} + 0.095163\,n_{\mathrm{ref},k-1}\\
c_k &= 0.999625\,c_{k-1} - 3.694\,972\times10^{-6}\,n_{k-1} + 3.147\,116\times10^{-5}\,(\dot m_{1,k-1}+\dot m_{2,k-1})\\
&\quad - 1.878\,394\times10^{-7}\,n_{\mathrm{ref},k-1} + 8.521\,345\times10^{-6}\,(\dot m_{\mathrm{ref},1,k-1}+\dot m_{\mathrm{ref},2,k-1})\\
&\quad - 4.999\,063\times10^{-5}\,R_{k-1} + 3.749\,297\times10^{-4}\,c_{\mathrm{in},k-1}\\
\dot m_{j,k} &= 0.606531\,\dot m_{j,k-1} + 0.393469\,\dot m_{\mathrm{ref},j,k-1}, \qquad \text{lines with } \sigma_j = 1\\
c_{m,k} &= 0.950110\,c_{m,k-1} - 9.449\,822\times10^{-8}\,n_{k-1} + 0.049880\,c_{k-1}\\
&\quad + 8.569\,178\times10^{-7}\,(\dot m_{1,k-1}+\dot m_{2,k-1}) - 3.189\,862\times10^{-9}\,n_{\mathrm{ref},k-1}\\
&\quad + 1.492\,579\times10^{-7}\,(\dot m_{\mathrm{ref},1,k-1}+\dot m_{\mathrm{ref},2,k-1}) - 1.257\,720\times10^{-6}\,R_{k-1}\\
&\quad + 9.432\,897\times10^{-6}\,c_{\mathrm{in},k-1}
\end{aligned}
$$

Lines with $\sigma_j = 0$ do not appear in the c equations.

Input–output

Each transfer function from Section 6.1 is discretized exactly with ZOH.

$$\frac{Y_1(z)}{U_1(z)} = \frac{\Delta h}{n_{\mathrm{ref}}} = \frac{2.042\,285\,28\times10^{-6}\,z + 1.954\,463\,26\times10^{-6}}{z^2 - 1.87347534\,z + 0.87645984} \tag{18}$$

$$\frac{Y_3(z)}{U_2(z)} = \frac{c_m}{\dot m_{\mathrm{ref},j}} = \frac{1.492\,578\,58\times10^{-7}\,z^2 + 5.224\,854\,70\times10^{-7}\,z + 1.133\,103\,18\times10^{-7}}{z^3 - 2.55626615\,z^2 + 2.13232854\,z - 0.57605504} \tag{19}$$

$$\frac{Y_3(z)}{D_1(z)} = \frac{c_m}{R} = \frac{-1.257\,719\,57\times10^{-6}\,z - 1.236\,291\,92\times10^{-6}}{z^2 - 1.94973549\,z + 0.94975419} \tag{20}$$

$$\frac{Y_3(z)}{U_1(z)} = \frac{c_m}{n_{\mathrm{ref}}} = \frac{-3.189\,861\,71\times10^{-9}\,z^2 - 1.228\,717\,97\times10^{-8}\,z - 2.957\,079\,36\times10^{-9}}{z^3 - 2.85457291\,z^2 + 2.71394782\,z - 0.85937313} \tag{21}$$

$$\frac{C(z)}{U_2(z)} = \frac{c}{\dot m_{\mathrm{ref},j}} = \frac{8.521\,344\,72\times10^{-6}\,z + 7.214\,478\,24\times10^{-6}}{z^2 - 1.60615573\,z + 0.60630325} \tag{22}$$

As difference equations:

$$
\begin{aligned}
y_{1,k} &= 1.87347534\,y_{1,k-1} - 0.87645984\,y_{1,k-2} + 2.042\,285\,28\times10^{-6}\,u_{1,k-1}\\
&\quad + 1.954\,463\,26\times10^{-6}\,u_{1,k-2}\\
y_{3,k} &= 2.55626615\,y_{3,k-1} - 2.13232854\,y_{3,k-2} + 0.57605504\,y_{3,k-3}\\
&\quad + 1.492\,578\,58\times10^{-7}\,u_{2,k-1} + 5.224\,854\,70\times10^{-7}\,u_{2,k-2} + 1.133\,103\,18\times10^{-7}\,u_{2,k-3}\\
y_{3,k} &= 1.94973549\,y_{3,k-1} - 0.94975419\,y_{3,k-2} - 1.257\,719\,57\times10^{-6}\,d_{1,k-1}\\
&\quad - 1.236\,291\,92\times10^{-6}\,d_{1,k-2}\\
y_{3,k} &= 2.85457291\,y_{3,k-1} - 2.71394782\,y_{3,k-2} + 0.85937313\,y_{3,k-3}\\
&\quad - 3.189\,861\,71\times10^{-9}\,u_{1,k-1} - 1.228\,717\,97\times10^{-8}\,u_{1,k-2} - 2.957\,079\,36\times10^{-9}\,u_{1,k-3}\\
c_k &= 1.60615573\,c_{k-1} - 0.60630325\,c_{k-2} + 8.521\,344\,72\times10^{-6}\,u_{2,k-1}\\
&\quad + 7.214\,478\,24\times10^{-6}\,u_{2,k-2}
\end{aligned}
$$

Check with the final value theorem, $y[\infty] = \lim_{z\to 1}(z-1)\,G(z)\frac{z}{z-1} = G(1)$: the steady-state gains are 1.339 mm/rpm, 0.1067, −0.1333 and −0.0104 for the four paths, the same as in continuous time.

The contributions from each input add up. For example, the full measured O_2 is $y_{3,k}$ = (response to u_2) + (response to u_3) + (response to d_1) + (response to u_1), each computed with its own equation above.

7 Control design and measurement chain

The controller and filter settings the checks below use are part of [TOML]; they are the values in Heimdall's `ConfigValues`.

Symbol	Value	Unit	Source	Reason
K_c	450	rpm/m	[HEIM] ConfigValues	LIC01 proportional gain (rpm of pump speed per metre over-height); [FD] P = 1 normalised (90 rpm per 0.20 m). Util.PID form $u = K_c(e + \frac{1}{T_i}\int e\,dt)$ [OGA] 2-3.
T_i	60	s	[HEIM] ConfigValues	LIC01 integral time, chosen for robustness (worst-case phase margin in the design basis section on LIC01, [OGA] 7-7); [FD] gives 600 s, about 20 times the process time constant.
T_d	0	s	[HEIM] ConfigValues	No derivative action (PI): the level reading is noisy and quantized.
f_c	0.05	Hz	[HEIM], [PRO] 10.3.3–10.3.4	Cut-off of the 2nd-order Butterworth low-pass (bilinear transform with pre-warping, SCL.Butterworth) on each level transmitter; ten times above the loop crossover, attenuates waves of 4–10 s period by about 19 dB.
θ	2	s	-	Dead time in the level loop: Modbus TCP poll, task cycle, OPC UA to the Framo PLC and the drive. The margins are also given for 2.5θ.[A]
T_{s,O_2}	24	s	[HEIM] ModbusRTUData	Time between two readings of the same Orbit 864: five sensors polled one at a time over the 750-652 (2–3 s each), the oxygen block every second round. The sensor lag τ_s acts as the anti-aliasing filter [PRO] 6.2.[A]

7.1 Level controller LIC01

LIC01 is a PI controller, $u = K_c(e + \frac{1}{T_i}\int e\,dt)$ ([OGA] §2-3), on the over-height. The integral action removes the steady-state error after a step disturbance, such as a change of the damper opening or of the outside level ([OGA] §5-7). The open loop is the PI, the pump lag, the level and the measurement filter, plus a dead time θ:

$$L(s) = K_c\Big(1 + \frac{1}{T_i s}\Big)\,\frac{K}{(1 + s/a_h)(1 + T_p s)}\,F(s)\,e^{-\theta s}, \qquad K = \frac{N_p K_n}{A_s a_h} \tag{23}$$

with $F(s)$ the 2nd-order Butterworth of Section 7.3. The dead time only lowers the phase, by $\omega\theta$, and so sets the upper limit for the gain ([OGA] §7-2, transport lag, and Example A-7-10). The margins per mode, at θ and 2.5θ ([OGA] §7-7; computed by [PY]):

					θ		2.5θ	
Mode	N_p	K [mm/rpm]	$1/a_h$ [s]	$1/\omega_c$ [s]	PM [°]	GM [dB]	PM [°]	GM [dB]
O-35	3	2.55	40	43	80	16.1	76	13.2
O-45	2	1.34	31	88	94	20.2	92	17.4
O-45	3	2.01	31	52	90	16.7	86	13.9
O-60	2	1.19	28	100	96	20.6	94	17.8
O-60	3	1.79	28	58	94	17.0	91	14.3
S-75	2	1.49	35	78	91	19.9	89	17.1

$1/\omega_c$ is the inverse gain-crossover frequency, a measure of the closed-loop time constant. The worst case is O-35 on three pumps, where the process gain is highest. [OGA] recommends a phase margin of 30–60° and a gain margin above 6 dB (p. 467). The chosen setting gives 76° and 13.2 dB in the worst case: deliberately on the safe side, because h_ρ, the real outlet conductance and the wave noise are uncertain. Faster settings stay inside the recommendation (worst case over the modes, dead time 2.5θ):

K_c [rpm/m]	T_i [s]	PM [°]	GM [dB]	$1/\omega_c$ [s]	
450	60	76	13.2	100	(set)
450	45	64	12.6	78	
600	45	55	10.1	56	
700	45	49	8.8	46	
450	600	120	14.5	943	

One fixed setting covers all modes and pump counts. The process gain grows with N_p, so a gain schedule would lower K_c with more pumps; [FD] raises it.

Implementation in Heimdall. LIC01 is `PIDController` around CODESYS `Util.PID` [UPID]:

- The integral part is held while the output is at its limits (40–130 rpm), which acts as anti-windup [UPID]. [OGA] does not treat windup.

- `Util.PID` applies new parameters only in manual mode, at a reset or at a start [UPID]. A new T_i takes effect at the next start or manual→auto switch.
- **Start-up tracking ([FD]).** While the pumps in level control are starting and none is in REMOTE, LIC01 runs in manual on their mean actual speed.
- **Bumpless transfer.** On manual→auto the PID is reset with the offset $Y_{\text{off}} = u_{k-1} - K_c e_k$, so the first automatic output equals the last manual one; the output is held in the reset cycle, where `Util.PID` returns Y_{off} alone.
- **Level measurement lost.** With no valid level pair (Section 7.3), `Pen.Status.LevelControlOK` is FALSE, the alarm `LevelMeasurementLost` is raised and LIC01 runs in manual at `PenConfig.LevelControl` (60 rpm on the test rig), limited to 40–130 rpm. When a pair is valid again, LIC01 returns to auto bumplessly. An operator in manual keeps the operator's output.
- **Heartbeat loss (HOLD, [FD] Tabell 5).** When the link to a pump in level control is lost (no heartbeat answer within `InletConfig.HeartbeatTimeout`; the PLC then reports the pump UNAVAILABLE and sends no start), LIC01 freezes its output. It stays frozen while the link is down and, once back, while the pump is in HOLD. When the pump is in REMOTE again, LIC01 runs one cycle on the mean actual speed of the pumps in level control and returns to auto bumplessly. The pump's `Status.EnableOxyControl` is TRUE only with a live link in REMOTE: the permissive for oxygenation on that line.

Test on 13 October 2026 (PFC200 against CasRioSim and CasFramoSim, three pumps, O-35 outlet, setpoint 6 cm, $K_c = 450$ rpm/m, $T_i = 60$ s, filter 0.05 Hz): the pumps reached REMOTE at 87 rpm with no jump in the setpoint; the level overshot to 7.4 cm, was within 2 mm of the setpoint about 3 min after the transfer, and settled at 74.3 rpm (model: 73.9 rpm). Stopping one of the three pumps gave a dip of 3.1 cm and a slow recovery, since two pumps need about 111 rpm; restarting it gave an overshoot above 7.4 cm.

Set-point kick. The P part acts on the error, so a setpoint step (e.g. a mode change from 6 to 0.5 cm) gives a step in pump speed, and at a manual→auto transfer the offset absorbs $K_c e$. The I-PD form, with the P part on the measurement and the setpoint only in the integral, avoids both and keeps the same disturbance response ([OGA] §8-5, Examples A-8-8 and A-8-9). `Util.PID` has no such form (open item).

7.2 Oxygen control in Heimdall

OIC01 works in total flow. OIC01 (`InletSupervisor`) is a PI from the pen O_2 in mg/L (mean of the sensors for control, `Pen.Status.PenOxygenLevel`) to the total O_2 flow to the pen, $\dot{m}_{\text{tot}}$ in kg/h. Since the dissolved O_2 depends on the total flow (Section 6), the loop gain does not change with the number of lines running, so the three parameter sets of [FD] Tabell 13 are not needed. Setpoint and tuning are retained in `PenConfig` (`OxySetpoint`, `OxyController`).

Sharing. Let A be the lines in auto (oxygenation active, FIC and valve in auto, good flow measurement) and M the active lines that are not (valve in manual, bad flow). Each line in auto gets

$$\dot{m}_{\text{sp},i} = \min\Big(\dot{m}_{\text{max},i},\ \frac{\dot{m}_{\text{tot}} - \sum_{j \in M} \dot{m}_{\text{meas},j}}{|A|}\Big), \qquad i \in A, \tag{24}$$

so a manual line counts with its *measured* flow. When a line drops out, or an operator moves a manual line, the lines in auto take over the difference in the same cycle; the change does not go through the integral of OIC01, and gives no over- or undershoot of its own. The output of OIC01 is limited to what the lines can deliver, $\sum_A \dot{m}_{\max,i} + \sum_M \dot{m}_{\text{meas},j}$, so the integral does not wind up beyond reach. It starts bumplessly from the measured total flow, and freezes while no O_2 sensor for control is good ([FD] §3.2.4).

Per line (`OxyControl`): the line starts on the operator's group start only while its inlet pump runs in REMOTE with a live link (`InletPump.Status.EnableOxyControl`); the oxy pump then starts in auto. PIC41–43 sets the pump speed (20–50 Hz) for 4.0 bar absolute after the pump; while starting, in pump manual or with a bad pressure, the pump runs at its preset speed (40 Hz). After the pump has run `RunDelay` (30 s), oxygen may be added: FIC41–43 sets the valve for the line's share; its output reaches the valve at most 0–100 % in 2 s (`ValveRampTime`), a rate limit in the PLC rather than in the valve. The valve output is held at 4 mA (closed) otherwise, so it never opens without the pump running. The ATV320 is driven over Modbus TCP on the second channel of its line's InletRIO master (`InletRIO.DriveChannel`, its own connection since the drive has its own IP): CMD/LFR written, ETA/RFR/LCR/THD/OPR read, DRIVECOM state machine in `OxyPumpATV320`.

Start values (`ConfigValues`, to be tuned at commissioning): OIC01 $K_c = 50$ (kg/h)/(mg/L), $T_i = 1800$ s (pen time constant about 44 min, about 0.024 mg/L per kg/h); FIC $K_c = 0.3$ %/(kg/h), $T_i = 5$ s; PIC $K_c = 2$ Hz/bar, $T_i = 5$ s.

7.3 Measurement chain

Sampling. The PLC reads the level transmitters every Modbus TCP cycle, about 0.1 s, so $F_s/2 \approx 5$ Hz. Waves of 4–10 s period (0.1–0.25 Hz) are far below that, and the sensors' own damping acts as the analog pre-filter, so aliasing is negligible ([PRO] §1.4.2, eq. 1.4.19; §6.2).

Quantization. The 750-455 resolves the 2 m span in 2048 steps, $\Delta = 0.98$ mm. Rounding gives an error within $\pm\Delta/2$ with power $\Delta^2/12$, i.e. $\sigma_q = \Delta/\sqrt{12} = 0.28$ mm ([PRO] §1.4.3–1.4.4, eqs. 1.4.26–1.4.29), well below the sensor deviation. That noise model assumes the signal moves across the steps ([PRO] §6.3.3). A noise-free, slowly drifting level does not; the PI then hunts across one step (a sawtooth of $\pm K_c\Delta \approx \pm 0.45$ rpm, seen in the simulator before it had sensor noise). The simulator therefore adds the sensor noise σ_{LT0x}, σ_{LT2x} ([TOML]), which dithers the quantizer as the real sensors will.

Filter. Each level pair's over-height Δh_i (inside minus outside, from the unfiltered scaled readings) has a 2nd-order Butterworth low-pass, $|H(\Omega)|^2 = 1/(1 + (\Omega/\Omega_c)^4)$ ([PRO] §10.3.4, eq. 10.3.45), with cut-off $f_c = 0.05$ Hz. It is made digital with the bilinear transform $s = \frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}$ and pre-warping of the cut-off, $\Omega_c = \frac{2}{T}\tan(\pi f_c T)$, so that the -3 dB point is exactly f_c ([PRO] §10.3.3, eqs. 10.3.40–10.3.43, Example 10.3.5). With $K = \tan(\pi f_c T)$ and $n = 1/(1 + \sqrt{2}K + K^2)$:

$$y_k = b_0 u_k + b_1 u_{k-1} + b_2 u_{k-2} - a_1 y_{k-1} - a_2 y_{k-2} \tag{25}$$

$$b_0 = b_2 = K^2 n, \quad b_1 = 2b_0, \quad a_1 = 2(K^2 - 1)n, \quad a_2 = (1 - \sqrt{2}K + K^2)n$$

- Heimdall implements it as `FB.Components.Butterworth2`, one section in direct form II transposed ([PRO] §9.3). With $f_c \ll F_s$ the poles lie close to $z = 1$, where finite-precision coefficients allow only a sparse grid of pole positions and the DC gain drifts ([PRO] §9.5, eq. 9.5.13), so coefficients and states are LREAL.

- The sample time T is the cycle time of the calling task, read once from the runtime (`GetTaskIntervalUs`, CmpIecTask), so the cut-off holds whatever the Task Configuration says. If the runtime gives none (an event task), the running mean of the measured call interval is used instead, with the coefficients recomputed when it drifts by more than 2 %. The per-transmitter display filter reads T the same way.
- When a pair becomes valid, and at the first call, the filter state is set to the steady state at the input ($s_2 = (b_2 - a_2)u$, $s_1 = (b_1 - a_1)u + s_2$, using DC gain 1), so a returning pair enters the PV at its value instead of ramping up from 0.
- Filtering the difference rather than each transmitter gives the same result while every transmitter is healthy (the filter is linear), but costs one filter per pair and restarts cleanly when a pair returns.
- The filter adds phase lag at the loop crossover, which the margins above include ($F(s)$). It attenuates 4–10 s waves by about 19 dB. Checked offline at $f_c = 0.05$ Hz, $T = 10$ ms: -3.01 dB at f_c; a 1 cm step reaches 63 % in 5.6 s with 4 % overshoot.
- `SCL.Butterworth` (WORD input) still smooths each transmitter for display; it is not in the control path. Its earlier version applied 2π twice and put the cut-off about 27 times too high, so it hardly filtered.

Oxygen measurement. The Orbit 864 sensors share one RS-485 line and are polled one at a time over the 750-652, the oxygen block every second round. The same sensor is read about every $T_{s,\mathrm{O_2}}$ ([TOML]), not every T_s. The sensor lag τ_s band-limits the signal well below the Nyquist frequency of that interval, acting as the anti-aliasing filter ([PRO] §6.2). OIC01 should be designed for that interval; against the pen time constant of 44 min it is short.

8 Simulators

Two simulators run this design basis, split along the PLC's real interfaces:

- **CasRioSim**: the four Wago remote I/O stations over Modbus TCP, and the pen physics (level, oxygen, valves, outlet, sensors);
- **CasFramoSim**: the three Framo inlet pump controllers over OPC UA (Section 8.2).

They share nothing but OPC UA and the design basis: CasRioSim's physics reads each pump's actual speed from its CasFramoSim server, as an OPC UA client. Both report the design basis version, 2.3.11.

8.1 Remote I/O and pen physics (CasRioSim)

CasRioSim serves the Heimdall PLC the same Modbus TCP process image as the four Wago remote I/O stations. The plant behind it is this design basis: it integrates the nonlinear equations (3)–(7) from [MODEL] with the parameters in [TOML], using RK4 at 0.1 s. The simulator version is the design basis version, 2.3.11; every station reports it in holding registers 2013–2015.

The simulator differs from the control design model only where the PLC, not the plant, owns the behaviour:

- **One speed per pump.** Each inlet has its own speed state. With CasFramoSim running, the speed is read from the pump's Framo server, which runs the lag $T_p \dot{n}_j = n_{\mathrm{ref},j} - n_j$ itself; otherwise CasRioSim integrates it from a setpoint in its control block. With a common speed this is equation (4).

- **O_2 valve instead of the closed FIC.** The PLC runs FIC41–43 and writes the valve position $u_{\%,j}$. The plant is the valve itself: $\tau_v \, \ddot{m}_j = \sigma_j \, \dot{m}_{\max} \, u_{\%,j}/100 - \dot{m}_j$, zero below the cut-off u_{cut}. FT41–43 read $\dot{m}_j$.

- **Outlet opening.** $q_{\mathrm{out}} = k_{d,\mathrm{full}} \, z \, \mathrm{sgn}(\Delta h + h_\rho) \sqrt{|\Delta h + h_\rho|}$ with opening $z \in [0, 1]$, set from the simulator control block or from the mean damper cylinder position. A negative q_{out} is sea water flowing in through the outlet; it carries c_{in}.

- **Start state.** Pumps stopped, $\Delta h = -h_\rho$, $c = c_{\mathrm{in}}$.

The self-test (`CasDesignBasis/tools/selftest_plant.py`) checks that every operating point in Section 5 is an equilibrium of the simulator plant, and that the plant linearizes to exactly **A**, **B** and **E** of Section 6 in every mode.

The cylinder counters, CO_2, total gas pressure and oxy-pump pressures are simulated instruments with static curves; they are not part of the design model.

O_2 sensors. The PLC reads the Orbit 864 sensors (In-Situ RDO Core) over Modbus RTU through the 750-652 serial module, the first module on CommonRIO, so its 24-byte process image is Modbus words 0–11 [W652]. CasRioSim runs that module's handshake on the words the PLC writes and reads with FC23 (initialization IR/IA, transmit TR/TA with length OL, receive RR/RA with length IL, up to 22 bytes per step), and behind it five Modbus RTU slaves: OT01–03 inside the pen (addresses 1–3) read c_m, OT21–22 outside (4–5) read c_{in} [HEIM]. Each answers function code 3 with the register map of a real Orbit 864, read from the sensor on the test rig on 7 October 2026. The sensor is an RDO Core IC: one sub-sensor per register block, each laid out like the oxygen block of [RDOM] (sensor id at index 0, parameters from index 37, 8 registers apart):

- oxygen, registers 40001–40066, sensor id 42: DO in mg/L as a float at 40038–39 (parameter Id 20, unit 117), temperature at 40046, saturation at 40054 (from the same saturation formula as c_{sat}) and O_2 partial pressure at 40062;

- conductivity, 40873–40957, sensor id 85: actual and specific conductivity, salinity, total dissolved solids, resistivity and water density at 40910, 40918, 40926, 40934, 40942 and 40950;

- pressure, 41745–41797, sensor id 84: pressure (psi) at 41782 and depth (m) at 41790.

[RDOM] documents only the oxygen block. Register 4xxxx is PDU address xxxx−1, and floats are sent high word first; both were confirmed with consistent readings (8.68 mg/L at 24.4 °C is 103.8 % saturation). The "PLC interface" of [RDOM] (register $5451 + 7(\mathrm{id} - 1)$) is refused by the sensor. The reply reaches the PLC after its wire time at 9600 baud 8N1 (as ScaleAQ configures the sensors). A sensor left idle sleeps and answers the request that wakes it about 3 s late, as seen on the test rig; register 9210 (Communication Stay Awake) keeps it awake. One poll of the oxygen block takes about nine Modbus TCP round trips, some 0.6 s. The module's register communication is not documented; the simulator answers it generically. The same value is in the simulator control block (holding register 2008, mg/L×100) for use without a PLC.

8.2 Framo inlet pump controllers (CasFramoSim)

Each inlet pump has its own Framo controller with an OPC UA server [SPEC]. CasFramoSim runs three identical servers, one container each, with the address space Heimdall's Data Source Manager binds to: the same names, numeric node ids and data types, in namespace `FramoPLS_1`–`3`. The node list is generated from the Data Source object (`framo_nodes.csv`, 305 nodes per pump); the ports are set per container (default 4841–4843, as in the Data Source endpoints).

Signal	Node (value)	Type	PLC
SpeedSetpoint	i=33 (`.Value` i=261)	AnalogValue [rpm]	writes
Start, Stop, Reset	i=30–32 (`.Active` 257–259)	Event	writes
HeartbeatToggle	i=34 (`.Active` i=277)	BOOL	sets (Framo clears)
MotorVelocity	i=2 (`.Value` i=59)	AnalogValue [rpm]	reads
MotorFrequency, Current, Power, Voltage, Torque	i=1, 3–6	AnalogValue	reads
PumpState… (7 states)	i=23–29 (`.Active` 250–256)	Event	reads
Alarms (11)	i=12–22 (`.Active`, `.Id`)	Alarm	reads
Simulation (fault injection)	i=40 (members 299–305)	BOOL / REAL	test only

The controller follows the state table in [SPEC] p. 6: UNAVAILABLE → AVAILABLE when no fault is active; Start (rising edge) → START, where the controller ramps the pump to n_{start}; at $n_{\text{start}} \pm \Delta n_{\text{start}}$ ("Sequence END") → REMOTE, where the pump follows SpeedSetpoint clamped to $n_{\min}$–$n_{\max}$. Stop → STOP, which ramps to zero and returns to AVAILABLE. A fault (DriveError, MotorTemperature, EmergencyStop, UPSSupplyError, DCUnderVoltageError) in START, REMOTE, HOLD or STOP → LOCKDOWN, which ramps to zero and ends in UNAVAILABLE.

Heartbeat. HeartbeatToggle is one Boolean that both sides toggle in turn: the PLC sets it while it is FALSE, the Framo controller clears it while it is TRUE, each an interval (1 s) after the other. The value always says whose turn it is, so after a (re)connect neither side waits for the other. Each side supervises only the other's half: the controller goes REMOTE → HOLD (start speed) when the PLC has not set the bit for T_{HB}, and back to REMOTE when it does; the PLC (`InletControl.Heartbeat`) raises its communication alarm when the controller has not cleared the bit for `InletConfig.HeartbeatTimeout`. The ramp is the plant's pump lag T_p, run in CasFramoSim; CasRioSim reads the resulting speed over OPC UA, so the simulated level follows the PLC's speed setpoint.

Drive feedback follows the SR2000 data: $f = nz_p/120$, $P = P_{\max}(n/n_{\max})^3$, torque P/ω (4407 Nm at 130 rpm, as on the datasheet), voltage $\propto f$, current scaled from power and voltage to 110 A at full speed. The evacuation (vacuum) nodes are present but static.

Symbol	Value	Unit	Source	Reason
n_{start}	88	rpm	[FD] 3.2.1	Speed the Framo controller ramps to after Start, before handing over in REMOTE ([FD]: "forhåndsinnstilt hastighet"). Set near n_0 in the nominal mode so the level is close to its setpoint at hand-over.[A]
Δn_{start}	1	rpm	-	START ends ("Sequence END") when the actual speed is within this of n_{start}; STOP and LOCKDOWN end below it.[A]
HB	on	–	[SPEC] p. 6	HeartbeatToggle is one Boolean that both sides toggle in turn: the PLC sets it, the Framo controller clears it. If true, REMOTE goes to HOLD when the PLC has not set it for the timeout; once the PLC sets it again, a Reset takes the pump back to REMOTE ([FD] Tabell 5). The simulator always clears the bit, also with supervision off. SIM_FRAMO_HEARTBEAT=0 turns supervision off.
$T_{\text{HB,int}}$	1	s	-	The Framo controller clears HeartbeatToggle this long after the PLC set it. Same as the PLC default InletConfig.HeartbeatInterval.[A]
T_{HB}	5	s	-	Heartbeat supervision timeout: longest time without the PLC setting HeartbeatToggle. More than two intervals plus the OPC UA update times. Same as the PLC default InletConfig.HeartbeatTimeout.[A]
$T_{\text{mot,trip}}$	135	°C	[SR2000]	Simulation.MotorTemperature above this raises the MotorTemperature alarm. Class B rise of 80 K over ambient, below the class F limit of 155 °C.[A]

Oxy pump drives (CasRioSim 2.3.11). Each inlet station also serves its line's oxy pump drive, a Schneider ATV320, over Modbus TCP (port 503; 5131–5133 on the test rig), as Heimdall's `OxyPumpATV320` talks to it: DRIVECOM state machine on CMD (8501) and ETA (3201), frequency reference LFR (8502) and output frequency RFR (3202) in 0.1 Hz, ramp t_{ramp}, and a communication fault (CnF) when the PLC stops writing. The station computes the absolute pressure after the pump, $p_0 + \Delta p_{\text{pump}}(f/f_{\max})^2$, and after the oxygenator, $p_0 + (\Delta p_{\text{pump}} - \Delta p_{\text{ox}})(f/f_{\max})^2$, and writes them to the PLC's pressure inputs. Below $f_{\text{ox,min}}$ no O_2 is dissolved on that line. The simulator UI injects a drive fault per pump. These curves are not part of the design model:

Symbol	Value	Unit	Source	Reason
$f_{\max}$	50	Hz	-	Drive output frequency at full speed (motor nominal; Heimdall OxyPump.MaxSpeed_Hz).[A]
p_0	1	bar	-	Absolute pressure at the transmitters with the pump stopped (atmosphere; static head neglected).[A]
Δp_{pump}	4.5	bar	-	Pressure rise of the oxy pump at $f_{\max}$; the rise goes with $(f/f_{\max})^2$ (affinity law, fixed system). 4.0 bar absolute after the pump (the PIC setpoint) then needs about 41 Hz, close to the 40 Hz preset speed.[A]
Δp_{ox}	1.3	bar	-	Pressure drop over the oxygenator at $f_{\max}$, also with $(f/f_{\max})^2$: PT after the oxygenator reads $p_0 + (\Delta p_{\text{pump}} - \Delta p_{\text{ox}})(f/f_{\max})^2$.[A]
t_{ramp}	5	s	-	Drive acceleration and deceleration ramp, 0 to $f_{\max}$ (ATV320 ACC/DEC).[A]
T_{CnF}	2	s	ATV320	Communication-loss fault (CnF): with the drive running, no write from the PLC for this long trips the drive.[A]
$f_{\text{ox,min}}$	10	Hz	-	Below this oxy pump speed no water flows through the oxygenator and no O_2 is dissolved on that line (simulator: the O_2 command of that line is zeroed).[A]

9 Design checks

9.1 Oxygen capacity

These are the loads on each oxygenator needed to hold c_{ref} in O-45, and the largest biomass the running lines can carry, for the cases in [TOML]:

		3 pumps		2 pumps	
Case	R [g/s]	Load [%]	Max. biomass	Load [%]	Max. biomass
10 °C, 1.0 kg (nominal)	54.3	88.0	1153 t	**132.1**	727 t
12 °C, 1.0 kg ([DB] dimensioning)	61.0	97.1	1032 t	**145.7**	653 t
16 °C, 1.5 kg	71.8	**112.0**	884 t	**168.0**	561 t

This is the most important finding. Each oxygenator delivers only while its own inlet pump runs. Consequences:

- **Normal operation:** 1 000 t is feasible only with all three pumps running, and with little margin: 3 % at 12 °C, and not enough at 16 °C/1.5 kg.
- **Redundancy case (one pump out):** [DB] covers the water flow with two pumps, but not the oxygen. Two oxygenators carry only about 560–730 t. Losing a pump at full biomass is

therefore an oxygen event, not only a flow event; alarms and fallback (emergency oxygenation, stopping feeding) must be designed for it.

To resolve this, do one of the following:

- size each oxygenator for $\approx 1.5\times$ the present load (about 150–175 kg/h), so that two lines carry full biomass;
- limit the biomass to what two lines can carry, about 650 t (33 kg/m^3) at 12 °C;
- accept the shortfall for the duration of a pump outage, with emergency oxygenation ([FD] §3.2.5) and stopped feeding (2.5 mg/(kg min) [O2S]) as the defined fallback.

9.2 Level loop

- The level time constant is 28–40 s with h_ρ, and 4–25 s without it. Whether h_ρ is included is the largest single uncertainty in the level model.
- With h_ρ, the outlet conductance k_d is almost the same in O-35 and O-45 (30.8 vs. 30.9). The [DB] level setpoints are thus consistent with one fixed opening plus the density head, which supports the assumption.
- The process gain grows with N_p. The controller gain should therefore fall as more pumps run ($K_c \propto 1/N_p$). The [FD] table has it the other way round (0.5/1/2 for 1/2/3 pumps). One fixed setting covers all modes (Section 7).
- The [FD] integral time $T_i = 600$ s is about $20\times$ the process time constant: a closed loop of about 16 min. LIC01 uses $T_i = 60$ s (Section 7); tested on the PFC200 against the simulators.
- The setpoint in O-60/S-75 is 5 mm, and the measurement uncertainty is about ± 6 mm (± 2 mm inside, ± 4 mm outside). With waves up to $H_s = 2$ m, this calls for heavy filtering, so the level loop cannot be fast.

9.3 Oxygen loop

- The pen behaves almost as an integrator (time constant 44 min) behind $\tau_s + T_f \approx 22$ s of lag, plus an unknown mixing delay θ.
- The [FD] FIC parameters ($K_c = 0.5$, $T_i = 600$ s) make the inner loop slower than the outer one, which defeats the cascade. The inner loop should have T_i of the order of 1 s.
- The coupling from n is negative and small, -0.010 (g/m^3)/rpm. A step of 10 rpm in the level loop moves O_2 by about -0.1 g/m^3. Feedforward from pump flow to $\dot{m}_{\mathrm{ref}}$ cancels it: $\Delta\dot{m} = -\frac{N_p K_n (c_{\mathrm{in}} - c_{\mathrm{ref}})}{\eta}\Delta n$.
- A feedforward base dose $\dot{m}_{\mathrm{ff}} = \frac{1}{N_o \eta}(rB - q_{\mathrm{in}}(c_{\mathrm{in}} - c_{\mathrm{ref}}))$ is what the [FD] review comment asks for (base oxygenation from fish count, size and temperature). It also gives a safe fallback if the O_2 sensor is lost.

10 Changes from CAS model.tex

Item	Old	New	Why
Capacitance C	1288 / 1886 m^2	2003 m^2	$D = 50.5\,\mathrm{m}$ [DB]. The old values match $r = 20.25$ and $r = 24.5\,\mathrm{m}$.
Outlet	linear, $K_{qo}/0.06$	$k_d\sqrt{\Delta h + h_\rho}$	Orifice law; k_d per mode from [FD]/[DB].
Outflow	fixed 9.52 m^3/s	per mode, 4.4–9.4 m^3/s	Mode table [FD] tab. 10.
O_2 washout	$K_{qo}x_2/V$	$q_{\mathrm{in}}(c_{\mathrm{in}} - c)$	Exact mixing balance. The old model counted the intake oxygen both as K_2u and through a fixed washout.
O_2 in intake	7 g/m^3 (100 % inflow)	7.27 g/m^3 (80 % sat.)	[O2S].
Fish consumption	500 mg/(kg h), 139 g/s	196 mg/(kg h), 54.3 g/s	Average post-smolt formula of [O2S] at 10 °C, 1 kg.
Valve gain	2.9 / 0.29 / 28.9	flow setpoint in g/s	Cascade [FD] rev. B; the valve sits inside FIC.
Valve time constant	0.02 / 2 s mixed	$T_f = 2\,\mathrm{s}$ (closed FIC)	One consistent value.
Transfer efficiency	1.0	0.8	[CTE].
O_2 sensor	not modelled	$\tau_s = 19.5\,\mathrm{s}$	[RDO].
Pump dynamics	none	$T_p = 10\,\mathrm{s}$[A]	To be confirmed by Framo.

11 Assumptions and open items

1. h_ρ **(density head).** This assumes the water inside, taken from 10–30 m, is 2 kg/m^3 denser than the surface water outside. Confirm against how [DB] computes the inner water height. Set $h_\rho = 0$ if the densities are equal.
2. T_p**.** Ramp and response of the Framo VSD and the water column, from setpoint to speed. Ask Framo [SPEC].
3. **O_2 setpoint.** OIC01 regulates concentration in mg/L (decided 13 October 2026); saturation is averaged for display only. [FD] does not yet give the value; c_{ref} at 90 % saturation is taken from [O2S]. Heimdall's OIC01 takes the retained `PenConfig.OxySetpoint` (default 8.0 mg/L, a placeholder until [FD] gives it).
4. **O_2 process value.** [FD]: OT03 at the outlet, falling back to the lowest remaining sensor. Heimdall: mean of the inside sensors in good quality.
5. **Mixing.** A fully mixed pen is assumed. The delay from injection to OT03 is not known; it needs CFD ([DB] §4.4.3) or a step test at commissioning.

6. η**.** The value 0.8 comes from a small-scale test (900 l, CT1). Confirm for CT10 at full scale.

7. **FS of LT21–23.** A full scale of 2 m is assumed for the accuracy.

8. **Direction of the level signal.** The level span is 2 m, which the 750-455 resolves in 2048 steps (0.98 mm per step). The VEGAPULS C 21 measures the distance from its antenna down to the water [VEGA]. Whether raw 0 (4 mA) is the lowest or the highest level depends on the min./max. adjustment and current output mode set in each sensor; the datasheets do not say. Read the settings in the sensors (VEGA Tools app). Inside (LT01–03) and outside (LT21–23) must have the same direction, otherwise the over-height, and the sign of LIC01, is wrong. The simulator assumes 0 = lowest level.

9. **Analog input word layout** (settled). The 750-455 puts the value in bits 14–4 (0–2047, 20 mA = `0x7FF0`); bit 15 is the sign, always 0 for 4–20 mA, and WAGO counts it in the stated 12-bit resolution. Bits 3–2 are reserved and bits 1–0 are diagnostics [W455]. WAGO Germany confirmed this layout on 7 October 2026 and called older tables (value in bits 14–3, 20 mA = `0x7FF8`) inconsistent. On a 2 m span one step is 0.98 mm, so the over-height, the difference of two readings, moves in steps of about 1 mm against level setpoints of 5 mm in O-60 and S-75. The simulator serves this layout. Heimdall's `RawMax` of 32760 reads 20 mA as 99.98 % of span.

10. **Quality check in the PLC** (fixed). `AnalogSensor.ScaleInput` takes a transmitter as healthy when status bits 1–0 are 00; the reserved bits 3–2 are not tested. To be verified on site at commissioning.

11. **Over-height** (fixed). Heimdall forms it per inlet pair and averages the valid pairs, as [FD] §3.2.2 asks (Section 3.6); a pair with a lost transmitter drops out, and with none left LIC01 runs at the safe speed. The simulator UI injects transmitter faults (wire break, over-range) to test it.

12. **Framo commands** (fixed). `InletControl` writes `InternalCommand` (Start, Stop, Reset, VelocitySetpoint).

13. **Framo heartbeat** (PLC code in review). Shared toggle bit, PLC sets and Framo clears (Section 8); the simulator's supervision is on and its UI injects a heartbeat loss. The PLC shows a pump UNAVAILABLE without a heartbeat answer and freezes LIC01 on heartbeat loss (Section 7). To confirm with Framo: that their controller clears the bit, and that the Data Source variable `Framo_n.HeartbeatToggle` is written by the PLC as well as read. The simulator returns from HOLD to REMOTE by itself when the heartbeat is back; [FD] Tabell 5 asks for an operator Reset.

14. **Copy-paste errors in the Framo data processing** (fixed).

15. **Framo interface vs. [SPEC].** The Data Source names differ from [SPEC] (SpeedSetpoint vs. VelocitySetpoint, DCUnderVoltageError vs. DCUndervoltageError, PowerConsumption-Counter vs. PowerConsumtionCounter); HeartbeatInterval and HeartbeatElapsed are not mapped; Limit*, DisplayName and Description are BOOL instead of REAL/STRING; each pump has its own namespace URI. Confirm against the delivered Framo PLCs. The simulator follows the Data Source.

16. **Speed setpoint at hand-over** (fixed). In START the PLC sends the pump's own speed, and LIC01 tracks it until REMOTE (Section 7).

17. **O_2 valve output scaling** (settled). The 750-563 in its factory setting (register 32 = `0x8016`) outputs 4–20 mA over the full 16-bit word: `0x0000` = 4 mA (closed), `0x7FFF` = 12 mA, `0xFFFF` = 20 mA (fully open). Heimdall's `AnalogValve.ScaleOutput` writes (Setpoint − Min)/(Max − Min) · 65535, limited to 0–65535, and the simulator decodes that range. After 100 ms without process data the module outputs 0 mA, below the cut-off, so the valve closes.

18. **Rigid surface.** The tarpaulin is flexible, so the real capacitance may be larger.

19. **Maximum flow.** [DB] gives 27 000 m^3/h as the maximum, while O-35 needs 34 000 m^3/h [FD]. The two documents must be aligned.

20. **Valve capacity.** A rough estimate puts one Bürkert 2865 (4 mm orifice) at about 57 kg/h of O_2. The nominal need is 92 kg/h per line (25.5 g/s), so check the valve against $\dot{m}_{\max}$. The Linde default setting of the 8605 driver (maximum 860 mA) is also above the 750 mA coil limit [B2865].

21. **Structure in the PLC function flow diagram** (fixed in Heimdall). Heimdall now has the [FD] rev. B cascade, OIC01 over FIC41–43, with OIC01 in total flow (Section 7.2). Align the old function flow diagram with it.

22. **Oxy pump pressure.** Heimdall takes InletRIO word 2 (`OxyPumpPressure`) as the pressure after the pump, into the oxygenator ([FD] PT41/43/45), and the PIC setpoint of 4.0 bar as absolute. Confirm the transmitters against the drawings; if they read gauge, set 3.0 bar.

23. **ATV320 over Modbus TCP.** To confirm at commissioning: FC23 on the drive's Ethernet port, unit id 0, command and reference channel on Ethernet (Fr1, Cd1), and the communication-loss fault (CnF) active, so the pump stops when the PLC is lost.

24. **Oxygen tuning.** OIC01, FIC41–43 and PIC41–43 run on start values (Section 7.2); tune at commissioning, OIC01 with a step test.

25. **O_2 sensor update rate.** The Orbit 864 delivers a value every 2–4 s, but polled on one RS-485 line the same sensor is read about every T_{s,O_2} (Section 7.3). OIC01 should not run faster than that.

26. **I-PD for LIC01.** P on the measurement instead of the error removes the set-point kick at mode changes and makes the transfer trivially bumpless ([OGA] §8-5). Needs a PI block of its own; `Util.PID` has only P on the error.

27. **Change of pump count.** A pump stopping or starting changes the flow per rpm, and LIC01 has to rebuild its output through the error (test: 3.1 cm dip, overshoot on restart). Sharing a total-flow output among the running pumps, with the contribution of a starting pump subtracted, would remove most of it. Deferred.

28. **Outlet dampers for flow control.** Planned: the hydraulic dampers become part of the flow control. The model then needs the damper characteristic (opening → k_d), the stroke time and the hydraulics.

29. **SCL.Butterworth version.** Only the per-transmitter display smoothing uses it now; check that the corrected filter is in the library version Heimdall references.